



MegaFlash for a2s4000 installation guide

Thank you for purchasing the MegaFlash for a2s4000 add-on feature for the original Apple IIc. The following document will assist you installing the device in your machine.

The MegaFlash has been shipped to you preconfigured with the following:

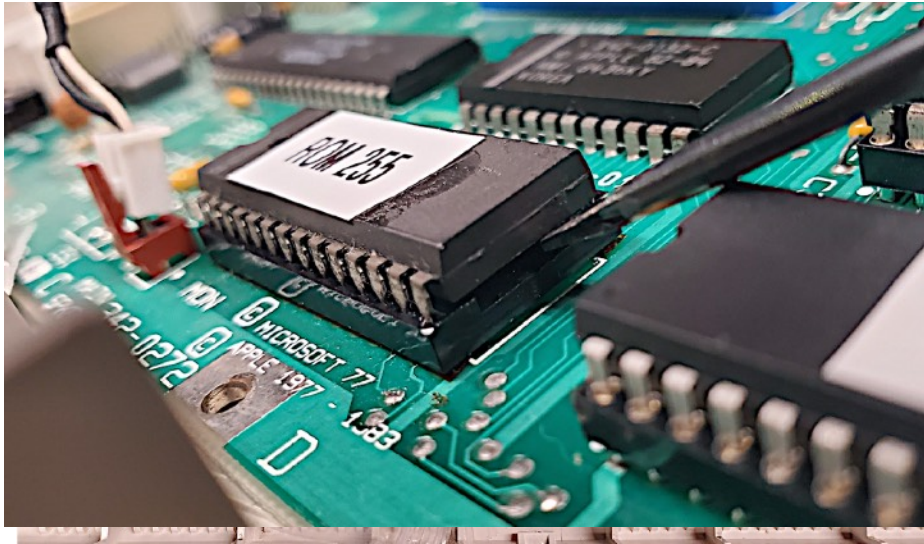
- Storage
 - Volume 1 - A2Desktop 1.5
 - Volume 2 - Total-Replay 5.3
 - Volume 3 - 32MB empty ProDOS volume
 - Volume 4 - 32MB empty ProDOS volume
- RAMDISK enabled (400KB)
- AppleSoft FPU enabled
- Latest release of MegaFlash firmware

Included in the package:

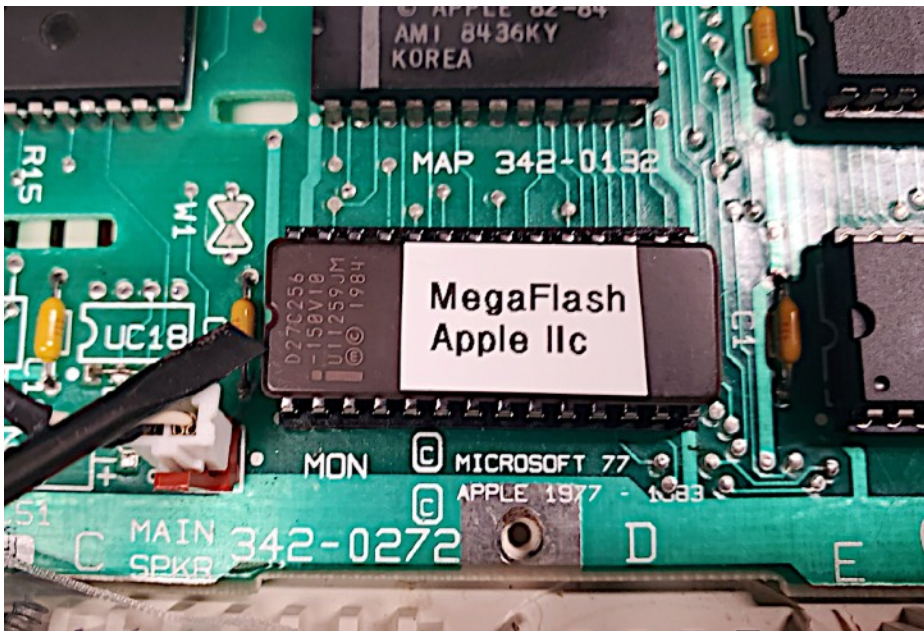
- MegaFlash for a2s4000 board
- MegaFlash ROM image for the Apple IIc
- Riser for the CPU socket
 - Flat pin riser to avoid damaging existing CPU socket

INSTALLATION INSTRUCTIONS

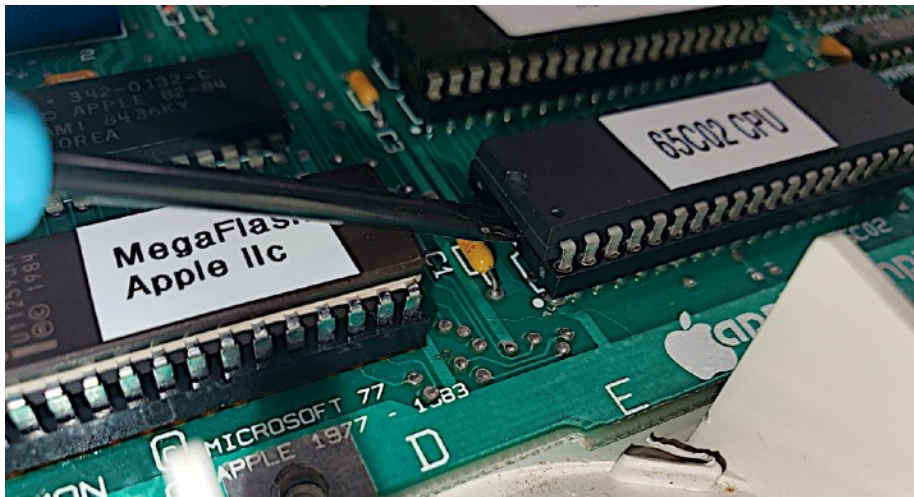
1. Disconnect your Apple IIc from power
2. Remove the cover from your Apple IIc. If unsure how, please search the web for videos appropriate for your model.
3. Once the top cover has been removed, lift away the keyboard. The pic below shows the board layout under the keyboard, with critical items labelled
4. With a small flat screwdriver, remove the MON ROM from the system, in this picture labeled as 'ROM 255'. Be careful to insert the screwdriver between the chip and the socket below, into the small gap between the two. Do not insert underneath them all, that may damage your board. Carefully lever the chip out of the socket, if necessary also lifting from the other side.



5. Insert the MegaFlash ROM into the empty MON ROM socket. Take extra care to ensure the chip's notch is towards the speaker connector. The label should be right side up.



6. Pin the old ROM to the foam previously attached to the MegaFlash ROM. Place the ROM back in the anti-static bag and save.
7. Remove the 65c02 CPU from the motherboard. Similar to removing the ROM, carefully insert the flat screwdriver between the CPU and socket. Lift only a little, then repeat the same from the other side. Repeat raising from one then the other side until the chip has been removed. Care is needed to ensure the pins do not get bent.



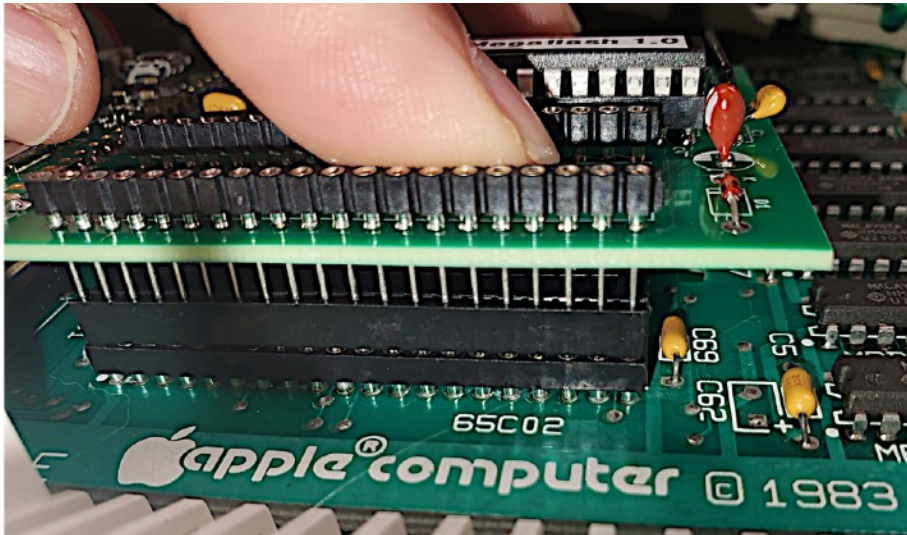
8. Insert a riser into the CPU socket.
 1. If this will not be a permanent install, please use the flat pin socket, as this will help ensure the CPU socket will still correctly mate with the original CPU if the MegaFlash is ever removed.
 2. For a more permanent and sturdy connection, use the round pin socket.
 3. Insert the selected riser into the CPU socket as shown below



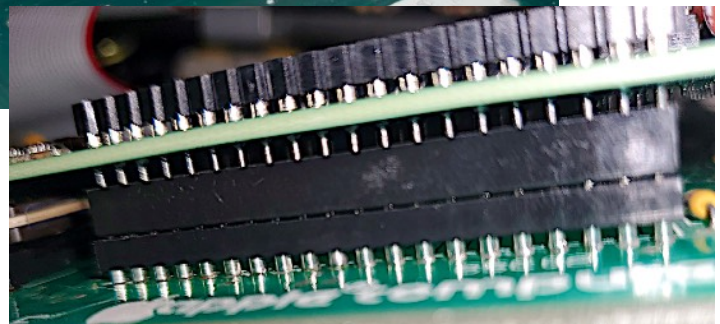
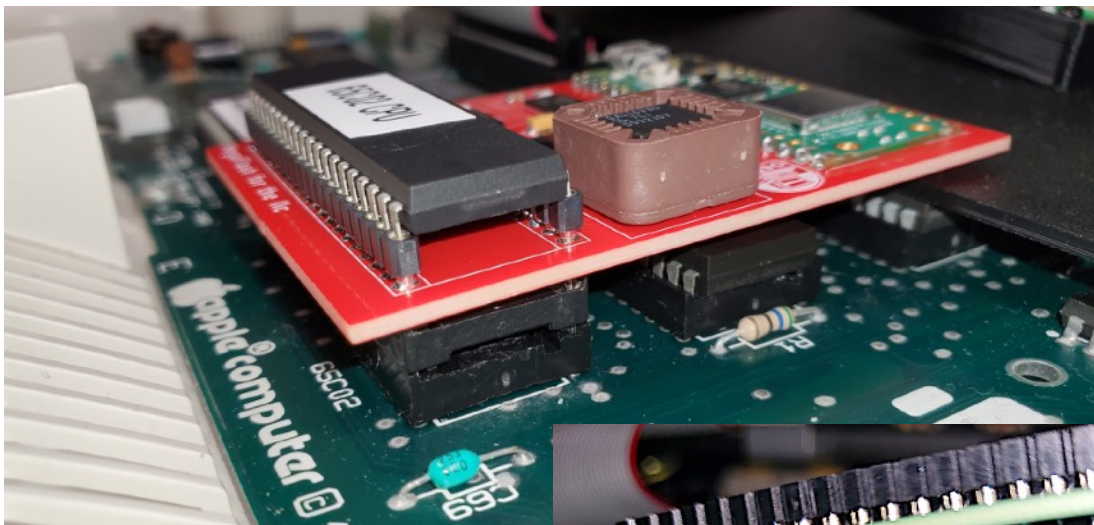
9. Insert the CPU into the empty socket on the MegaFlash with the protective pink anti-static foam still attached to the MegaFlash pins. Make sure the CPU notch is toward the left of the board, as in the picture below. The notch is also pictured on the board itself, for easier orientation. It may take a bit of jiggling to get the CPU seated in the socket, but be careful not to bend any CPU pins. Double check to ensure all pins are properly inserted into the holes, none of the pins are bent.



10. Before attaching the MegaFlash to the motherboard, remove the anti-static foam from the pins on the underside of the board. With the Pico to the rear or the machine, carefully align the MegaFlash pins to the CPU riser socket. Press firmly on top of the CPU to seat it into the socket. Quite a bit of force may be needed.



11. Properly seated into the socket, only a mm or two of pin should be showing



12. Congratulations! The board has been installed! You may now close your Apple IIc



QUICK-START USER GUIDE FOR THE MEGAFLASH

With the MegaFlash installed, your Apple IIc will immediately boot into A2Desktop, a GUI file management environment for the Apple II line of computers. On the desktop you will see the A2Desktop 32MB volume, Total Replay, two empty volumes (Vol3 and Vol4), and a RAM disk. If you have a floppy in your drive or external drives attached, these will also show.

CONTROL PANEL QUICK START

To access the control panel, press and hold the Closed Apple key to the right of the keyboard space bar. Simultaneously press Ctrl-Reset and release. Keep holding down the Open Apple key until the boot selector panel appears on the monitor. Press 7 to enter the control panel.

By default all features will be enabled, except WiFi and clock.

1. With the down arrow, Scroll down to TimeZone and set for your location using the left/right arrows
2. Click twice more with the down arrow to select 'WiFi settings'. Press the Enter key
3. Enter your wifi network name and press enter
4. Enter your wifi password and press enter
5. Back in the control panel, click the down arrow one more time to select 'Test wifi/NTP'. Press Enter
6. If you correctly entered your WiFi credentials the test will show the IP assigned to the MegaFlash and confirm Network Time functionality. If the connection fails, return to the WiFi setup and re-enter your wifi name and credentials.
7. Once the WiFi connection is complete, scroll down to 'Save and Reboot'. This will reboot the machine and boot into A2Desktop. The top right corner of the screen will show the current time.
8. Basic setup is complete.

BOOTING FROM DRIVE 2

IN#4 command to boot from drive 2 of MegaFlash

CONTROL PANEL DETAIL

DRIVES ENABLE

1. The flash drives and the RAM Disk can be individually enable/disable. Select Drives Enable from the main menu to open the window. Press the number keys to enable/disable the drives.
2. To save the changes, press Enter to return to main menu. Then, select Save and Reboot. To discard any changes, press Esc key.
3. Warning: The slot/drive mappings will be changed after drives are enabled/disabled.
4. Always identify the drives with ProDOS volume name instead of slot/drive number.

FORMAT

The MegaFlash drives and RAMdrive can be formatted from MegaFlash control panel. Two key features.

1. It can format drives to any arbitrary size.
2. It can erase the entire drive before formatting. It may take up to 2 minutes to erase a flash drive. Subsequent write access will be much faster if the drive is already erased.

MEGAFLASH HARDWARE/FIRMWARE INFORMATION

To show the hardware and firmware information, go to the boot menu. Hold the Open-Apple key while entering Control Panel. i.e. OA-7 or OA-C.

- Hold Open-Apple key until the device information screen appears.

ROM DISK

A ROM Disk is provided for disaster recovery. The ROM disk contains Copy II+ and ADT Pro. When enabled from boot menu, it will become the first unit of MegaFlash (slot 4, drive 1). Please note that the slot/drive assignments of all MegaFlash units are changed.

Ctrl-OA-Reset cold boot to disable the ROM Disk

RAM DISK

The RAM Disk can be enabled from Control Panel. The size of RAM Disk is 400kB. The RAM Disk is the last unit of SmartPort. Its volume name is /RAMDISK.

RAM Disk can be used to transfer floppy disk image with ADT Pro. Users may transfer a floppy disk image and write it to the RAM Disk. Then, copy files from the disk image to Flash storage.

APPLESOFT BASIC FPU

High bit of address \$5FC is the Enable flag of Applesoft FPU.

REAL TIME CLOCK AND NETWORK TIME SYNC

The Raspberry Pi Pico has a real time clock (RTC). But it does not support battery backup. MegaFlash gets the time from NTP server via WIFI.

To enable network time sync:

1. Configure the WIFI Settings from Control Panel
2. Run Test Wifi/NTP from Control Panel
3. If all tests are passed, enable the Network Time Sync option and setup the time zone.

If MegaFlash can fetch the time from NTP server, the time is shown at the bottom

right corner in Boot Menu and Control Panel.

If Network Time Sync option is enabled, a ProDOS clock driver will be installed automatically. The clock driver is in the patched system ROM. No software installation is required.

SLINKY EMULATION

With the stock Apple IIc firmware, MegaFlash can emulate a 256kB Memory Expansion card (aka Slinky) if Pico 2 controller board is used.

DISK IMAGE TRANSFER

Disk Image File (.po or .hdv) can be transferred to MegaFlash in 3 ways.

1. ADT Pro and Serial Port
2. MegaFlash Terminal Mode
3. Via WIFI and TFTP Server

TERMINAL MODE

To enter terminal mode, TURN OFF Apple IIc. Then, connect a PC to the Pi Pico USB port. Then connect to Pico with a Terminal Program like TeraTerm (Windows).

In the terminal mode, ProDOS-ordered disk image (.po or .hdv) can be uploaded to or downloaded from MegaFlash using XMODE-CRC or XMODEM/1K protocol.

DISK IMAGE TRANSFER VIA WIFI

Disk Image can be transferred between MegaFlash and TFTP server via WIFI. Note that TFTP is not FTP. TFTP protocol is much simpler and without user authentication.

To transfer a disk image:

1. setup the WIFI network from Control Panel.
2. Select Disk Image Transfer via WIFI from Control Panel and follow the instruction.

Please note that the TFTP server and MegaFlash must be on the same subnet and TFTP traffic cannot pass through firewall or NAT. In other words, MegaFlash and the server must be connected to the same WIFI router.

The block size of original TFTP protocol is 512 bytes. Later TFTP option extension allows larger block size. A larger block size increase transfer speed dramatically.

- Make sure blocksize option negotiation is enabled when configuring the TFTP server.

On Windows, Tfpd64 (<https://pjo2.github.io/tftpd64/>) server program is recommended. It has a GUI. You may need to run the app as administrator to save the setting. When sending a file to server, the file is overwritten without any warning if the file already exists on the server. Make sure Option negotiation is enabled.

On MacOS, consider using Pumpkin <https://github.com/mterrorn/pumpkin>

APPLE II DESKTOP

By default, Apple II Desktop copies itself to RAM Disk during start-up. This is not necessary for MegaFlash. This behavior can be disabled from Apple II Desktop Control Panel. Select Control Panel from Apple Menu. Open Options, then disable Copy to RAMCard.

UPDATING PI PICO BOARD FIRMWARE

MegaFlash by Osis uses the Raspberry Pi Pico 2W exclusively. Your board was delivered with the 1.1.7 firmware developed by Thomas Fok. Updates firmware from Thomas Fok is available here: <https://github.com/ThomasFok/MegaFlash/releases>

Alternate firmware is under development by Osis. This MegaFlash fork is working on improving the networking stack. It is not as mature as Thomas' version. Use with caution. The current release has a completely rewritten network stack, significantly improving TFTP file transfers. UtherNet II emulation is in development. The current release is available here: <https://github.com/eosis/MegaFlash/releases>

To upgrade the firmware:

1. Remove MegaFlash from your Apple IIc computer.
2. Press and hold the white button labeled BOOTSEL on the Pico board
3. Connect the Pico board to a USB port of a computer while holding the button. Wait a few seconds, then release the white button
4. A disk drive called RP2350 or RPI-RP2 should be mounted on your computer desktop.
5. Drag-and-drop the megaflash-pico2.uf2 firmware file downloaded from GitHub to the RP2350 drive. Wait until the file copy finishes.
6. Reinstall MegaFlash back to your Apple IIc computer carefully.